Cedar Falls Water Treatment Plant

1 Design Basis

The Cedar Falls Water Treatment Plant is sized for a design flow of 21.4 MGD (average day) and a peak flow of 33.1 MGD on the maximum day, serving a population of 79,355 persons.

Raw water turbidity reaches 9.1 NTU (95th percentile), measured over the last five years.

2 Treatment Process

The coagulation tank provides a detention time of 24 s at design flow, with a velocity gradient of 898 1/s. A sludge pump moves 247 gpm from the basins. Slow mixing follows in 3 stages, for a total detention time of 31.8 min at design flow.

There are 2 settling basins, each 151 ft long and 46 ft wide, with a side water depth of 14.1 ft; the surface loading rate is 0.58 gpm/ft² at design flow.

Filtration uses 9 dual-media filters of 625 ft² each. With one filter out of service, the filtration rate is 2.2 gpm/ft². The filters are backwashed at 18.1 gpm/ft².

Table 1. Low-lift pumps

Tag	Capacity (gpm)	TDH (ft)	Motor (hp)
LL-1	8,720	127.1	173
LL-2	4,646	95.7	278
LL-3	8,508	61.2	202

3 Chemical Feed

Alum is dosed at 40.2 mg/L on the average day and up to 76.1 mg/L on the maximum day. Sodium hypochlorite is fed at 4.2 mg/L ahead of the clearwell.

4 Disinfection Options

One option was considered for primary disinfection.

Table 2. Disinfection options

	Ultraviolet reactors
Capital cost ($M)	5.09
Annual energy use (MWh/yr)	669
Contact time at peak flow (min)	n/a